

MECHATRONIC SYSTEMS DESIGN LAB

Exercise 2

1. Use the switch 7 to rotate the servo in clockwise direction and switch 8 for counter-clockwise direction from initial position of servo (90 deg). Every single switch press and release event should correspond to 1 degree rotation.
2. Test the proximity sensor provided to you.
3. Write a code to Turn-ON a LED when the proximity sensor senses an obstacle and vice-versa.
4. Write a code to stop a servo motor (which is already sweeping between its maximum and minimum range) when the proximity sensor senses an obstacle and continue sweeping when the obstacle is removed.
5. Write a code to start/stop servo sweeping by receiving commands (Ex: 1- ON, 2-OFF) serially and wait till the user enters a valid command.

Note: Do nothing when an invalid input is entered. Brief instruction may be provided every time before the user enters the input.

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